

Xiaoyu Tong

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Education

PhD	University of Amsterdam , Natural Language Processing - Thesis: <i>Dissecting Incongruity: Metaphor and Humor Understanding of Large Language Models</i> - Supervisors: Dr Ekaterina Shutova, Dr Martha Lewis	Amsterdam, NL July 2026
MSc	University of Amsterdam , Brain and Cognitive Sciences - Thesis: <i>Metaphor Paraphrasing and Word-Sense Disambiguation: Toward a New Approach to Automated Metaphor Processing</i> - Supervisors: Dr Martha Lewis, Dr Ekaterina Shutova	Amsterdam, NL Mar 2021
MA	Xiamen University , Linguistics - Thesis: <i>The Writer's Operation of Extended Metaphor in Meaning Construction</i> - Supervisor: Prof. Zhiying Xin	Xiamen, China July 2017
BA	Xiamen University , English Minor in Chinese Linguistics	Xiamen, China July 2014

Experience

Independent Researcher , Independent Researcher - Research on LLMs' cultural alignment in humour processing, collaboration with Dr Ekaterina Shutova and Dr Martha Lewis at University of Amsterdam - Research on application of LLMs in speech-language therapy for people with aphasia, collaboration with Dr Jie Fu at Xingtai Medical College	Remote July 2026 – present 3 months
Faculty of Science, University of Amsterdam , Doctoral Researcher Conducted interdisciplinary research on evaluating LLMs' metaphor, humour, pragmatic, and cultural understanding in text and multimodal communication. - Developed theory-driven datasets, benchmarks, and evaluation protocols for testing the linguistic, pragmatic, and cultural capabilities in unimodal and multimodal LLMs. - Designed cognitively and linguistically grounded annotation schemes for figurative, humorous, and multimodal data; led annotator training and quality assurance, achieving high inter-annotator agreements across datasets. - Designed, implemented, and coordinated cross-cultural behavioural studies involving 42K human responses across 4K textual and multimodal stimuli. - Evaluated frontier open-weight and proprietary models using prompting and fine-tuning approaches. - Integrated statistical modelling, error analysis, and qualitative interpretation to characterise human-model alignment, identify systematic capability gaps, and explain variation in model performance.	Amsterdam, NL Sept 2021 – July 2026 4 years 11 months

Institute for Logic, Language and Computation, University of Amsterdam, Research Intern

Amsterdam, NL
May 2020 – Mar 2021
11 months

- Reframed a widely used, theory-driven manual protocol for metaphor identification (MIPVU) as a word-sense disambiguation (WSD) problem, and operationalised this insight in a novel dataset and evaluation framework for automated metaphor interpretation.
- Created a resource that led to the BIG-bench metaphor-understanding benchmark and MUNCH project.

Funding

- China Scholarship Council Doctoral Scholarship (2021 – 2025)

Teaching and Mentoring

- Thesis Mentor, MSc Logic, University of Amsterdam (2023)
- Teaching Assistant, Introduction to the Philosophy of Language, MSc Logic, University of Amsterdam (2022)
- Teaching Assistant, Introduction to Linguistics, BA English, Xiamen University (2015 – 2017)

Service

- Reviewer, ACL Rolling Review (2024 – present)
- Reviewer, *IEEE Transactions on Audio, Speech and Language Processing* (2026)
- Reviewer, *Journal of Cognitive Psychology* (2025)
- Reviewer, NAACL 2024 FigLang Workshop (2024)
- Member, Institute for Logic, Language and Computation (ILLC) PhD Council (2022 – 2023)

Publications

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|---|---------|
| Evaluating LLMs' cultural alignment in humor appreciation
Xiaoyu Tong , Ivo Verhoeven, Martha Lewis, Ekaterina Shutova
(In Preparation) | present |
| Dissecting Incongruity: Metaphor and Humor Understanding of Large Language Models
Xiaoyu Tong
hdl.handle.net/11245.1/0bfb1d70-c5e8-4538-a554-d4c0e2ec0980 (University of Amsterdam) | 2026 |
| Hummus: A Dataset of Humorous Multimodal Metaphor Use
Xiaoyu Tong , Zhi Zhang, Pia Sommerauer, Martha Lewis, Ekaterina Shutova
10.48550/arXiv.2504.02983 (Conditionally accepted by TACL) | 2026 |
| A framework for annotating and modelling intentions behind metaphor use
Gianluca Michelli, Xiaoyu Tong (co-first author), Ekaterina Shutova
10.18653/v1/2026.starsem-conference.4 (*SEM 2026) | 2026 |
| Metaphor Understanding Challenge Dataset for LLMs
Xiaoyu Tong , Rochelle Choenni, Martha Lewis, Ekaterina Shutova
10.18653/v1/2024.acl-long.193 (ACL 2024) | 2024 |
| Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models
Aarohi Srivastava, Abhinav Rastogi, ... Xiaoyu Tong , ... Ziyi Wu
openreview.net/forum?id=uyTL5Bvosj (TMLR, Featured) | 2023 |

Recent advances in neural metaphor processing: A linguistic, cognitive and social perspective 2021
Xiaoyu Tong, Ekaterina Shutova, Martha Lewis
[10.18653/v1/2021.naacl-main.372](https://arxiv.org/abs/10.18653/v1/2021.naacl-main.372) (NAACL 2021)

Projects

Hummus: Dataset and benchmark of humorous multimodal metaphor use 2026
github.com/xiaoyuisrain/humorous-multimodal-metaphor-use

MUNCH: Metaphor Understanding Challenge Dataset 2024
github.com/xiaoyuisrain/metaphor-understanding-challenge

Co-created dataset of intentions behind metaphor use 2024
github.com/GMichelli/intentions-behind-metaphor

BIG-bench metaphor understanding benchmark task 2021
github.com/google/BIG-bench/tree/main/bigbench/benchmark_tasks/metaphor_understanding

Selected Presentations

- Invited talk on Hummus at [Large Language METAPHORS Online Workshop](#), IUSS Pavia (Mar 2026)
- Research featured in UvA News, 'AI still struggles to get the joke' (June 2026)

Skills

Programming and research computing

LLM and VLM research

Dataset creation and research data management

Experimental design, data collection, and annotation

Data processing and statistical analysis

Research communication and publication

Languages

Mandarin Chinese
Native

English
Full professional proficiency, academic working language

Japanese
Proficient reading; strong conversational listening and speaking; limited formal writing

Interests

Artificial Intelligence

Linguistics

Cognitive Science